

Q1) Jai

a) 64 — (1)

b) Naive Bayes — (1)

NB-assumes Independence
directly calc. prob. in
single pass. $O(n \cdot d)$

CART — calculates Gini / Info
Gain at every split
for every feature. $O(n^2 \cdot d)$

$O(d \cdot n \log n)$ — (0.5)

c) $K=1$ — (1)

Training data will
have the point itself
as the nearest neighbour
— hood

— (1)

the counts
will be updated
to zero after
query resolution

Q2

(1.5)
← IG calc
right split

High Latency (5)(1+2)

right split

IG calculation
= 0.5 each * 4

Temp / Meetings / Deadline

Mild

High
Productivity

Hot

(1.5)(0.5+1)
Meetings / Deadline

Few

High

Many

Deadline

Yes

High

No

Low

Temp / Meetings

Hot

Low

Mild

Meeting

Many

Low

Few

High

(1.5)(0.5+1)

right split
IG calc

(1.5)(1.5)

1st split:

$\log_2$

IG values:

Meetings ≈ 0.085

Deadline ≈ 0.028

Temp ≈ 0.085

Caffeine ≈ 0.43

2nd split High Caffeine

Temp = Meetings = Deadline

IG ≈ 0.138

2nd split Low Caffeine

IG Temp = Meetings = 0.198

IG Deadline = 0.128

Q3. TA: Medha

a. False (F)

b. False (F)

c. True (T)

d. False (F)

e. True (T)

$\left[\begin{array}{l} +1 \rightarrow \text{correct} \\ -1 \rightarrow \text{incorrect} \end{array} \right]$